FIFTH SEMESTER B.Sc. DEGREE EXAMINATION, NOVEMBER 2018

(CUCBCSS—UG)

Physics/Applied Physics

PHY 5B 08/APY 5B 09—PHYSICAL OPTICS AND MODERN OPTICS

Time : Three Hours Maximum : 80 Marks

Section A

Answer in a word **or** *phrase.*

Answer **all** *questions. Each question carries* **1** *mark.*

1. To determine optical rotation of solutions ———— is used.

   (a) Interferometer. (b) Spectrometer.

   (c) Magnetometer. (d) Polarimeter.

2. Wide separation of spectral lines indicates :

   (a) High resolving power. (b) High magnifying power.

   (c) High dispersive power. (d) None of these.

3. A biprism produce N1 fringes with light of wavelength $\lambda 1$.The number of fringes with wavelength $\lambda 2$ is ————.

   (a) $\lambda 1N1/\lambda 2$. (b) $\lambda 2N1/\lambda 1$.

   (c) $\lambda 1/\lambda 2N1$. (d) $[\lambda 1/\lambda 2]^{1/2} N1$.

4. To invert a circularly polarized light, we use:

   (a) Quarter wave plate. (b) Halfwave plate.

   (c) Half shade device . (d) Biquartz device.

5. When focal length of a lens decreases, the field of view of the lens ————.

6. Colours of thin films are explained using the phenomenon called ————.

7. Which one is not used to produce interference pattern

   (a) Biprism . (b) Two mirror.

   (c) Newton's rings. (d) Nicol prism

8. Compact disc shows colours in white light due to ————.
9. The radius of the 6th zone plate of focal length 10cm for light of wavelength 6000 is ————.
10. The basic principle of optic fiber is ————.

(10 × 1 = 10 marks)

Section B

(Answer in a short paragraph- three **or** *four sentences.*
Answer **all** *questions. Each question carries 2 marks.*

11. Draw Frenel's two mirror arrangement ?
12. What is polarizing angle. How is it related to refractive index of the medium ?
13. What do you meant by double refraction ?
14. Define Translation matrix and refraction Matrix of an optical system.
15. Compare a zone plate and a convex lens ?
16. What are Newton's rings ? What are its uses ?
17. How are the hologram classified ? Explain.

(7× 2 = 14 marks)

Section C

Answer in **a paragraph** *of about* **half a page to one page.**
Answer any **five** *questions. Each question carries 4 marks.*

18. State and explain Fermat's principle of extremum path and use it to deduce the laws of reflection and refraction of light.
19. What is system matrix ? Obtain it in the case of a system of two thin lenses separated by a distance and hence derive the formula for its focal length.
20. What is meant by interference of light ? State the fundamental conditions for the production of interference fringes.
21. State the principle of superposition of light waves.
22. Define and explain polarization.
23. Distinguish between positive uniaxial and negative uniaxial crystals.
24. What is meant by critical angle of an optical fiber ? Obtain an expression for the critical angle.

(5 × 4 = 20 marks)

Section D

(Problems- write all relevant formulas. All important steps carry **separate** *marks.*
Answer any **four** *questions. Each question carries* 4 *marks.*

25. A soap film 5×10^{-5} cm thick is viewed at an angle of 35° to the normal. Find the wavelength of light in the visible spectrum which will be absent from the reflected light. Given $\mu = 1.33$?

26. What is the radius of the 1st half period zone in a zone plate behavior like a convex lens of focal length60cm for light of wavelength 6000Å ?

27. Calculate the thickness of (i) quarter wave plate(ii) a half wave plate. Given that $\mu_0 = 1.973$, $\mu_e = 2.656$ and $\lambda = 590$nm.

28. Show that areas of half period zones are equal.

29. Calculate the angle of acceptance of a given optical fiber, if the refractive indices of the core and the cladding are 1.563 and 1.498 respectively.

30. The critical angle of incidence for total reflection in case of water is 48°. What is its polarization angle ? What is the angle of refraction corresponding to the polarization angle ?

31. In Fraunhoffer diffraction pattern due to a narrow slit a screen is placed 2m away from the lens to obtain the pattern. If the slit width is o.2 mm and the first minimum lie 5mm on either sides of the central maximum, find the wavelength of light.

(4 × 4 = 16 marks)

Section E

Essays. Answer in about **two pages.**
Answer any **two** *questions. Each question carries* 10 *marks.*

32. Discuss the Fraunhoffer diffraction pattern due to a single slit. Draw the intensity distribution ?

33. Write principle of holography ? Describe the recording and rerecording process in holography with the help of suitable diagrams.

34. What are Newtons ring ? derive an expression for the radius of the nth dark ring formed by the reflection also by transmitted light.

35. Explain the rectangular propagation of light on the basis of Fresnels half period zone.

(2 × 10 = 20 marks)